Service Bulletin

Service Bulletin Number: 4383630	Released Date: 03-sep-2014
Ignition Coil Malfunctions	

Ignition Coil Malfunctions

This document provides guidelines for the reuse and replacement of ignition coils. Use this information in conjunction with the content provided in the corresponding service manual.

There are two types of ignition coils.

- Coil on plug: Feature ignition coil and coil extension.
- Coil beside plug: Feature ignition coil and spark plug wire with coil extension.

Figure 1, Coil on Plug

1. Ignition coil
2. Coil extension
3. Spark plug boot

Figure 2, Coil Beside Plug.

1. Ignition coil
2. Coil extension
3. Spark plug wire
4. Spark plug boot

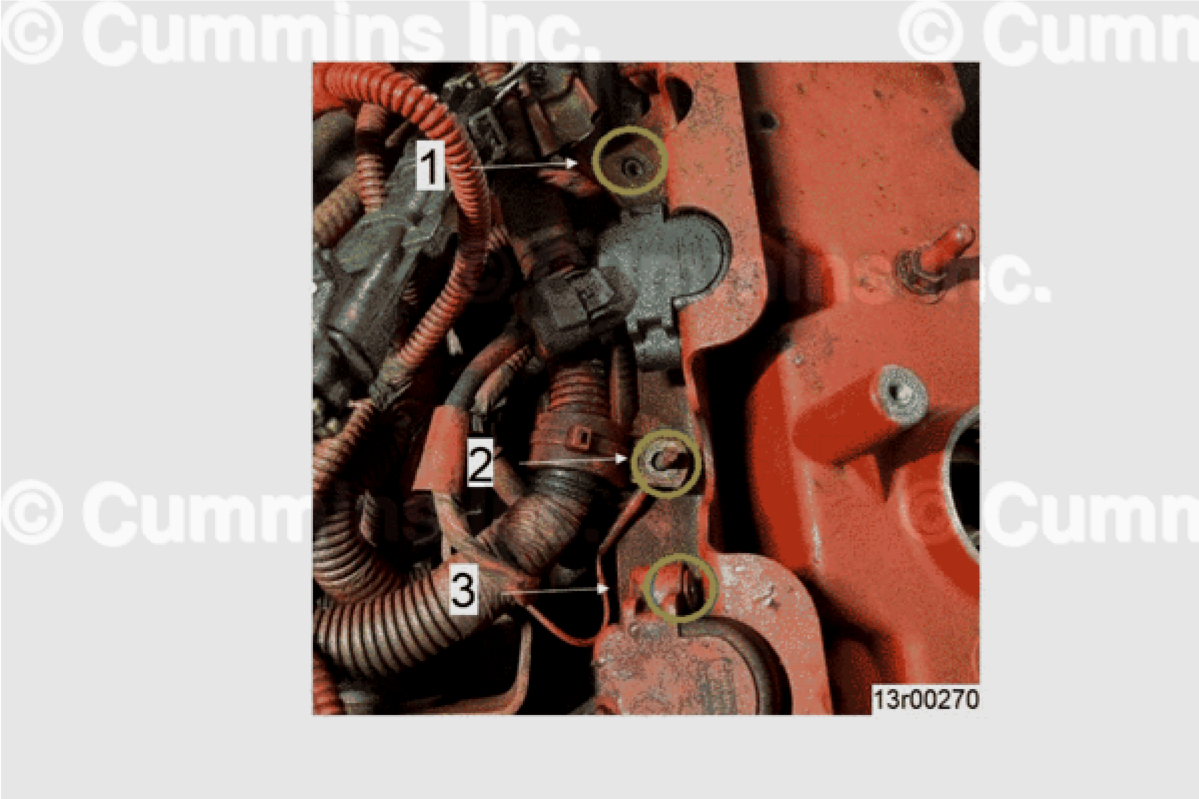

Figure 3.

Figure 3, Ignition Coil and Bracket Workmanship Allowing Ignition Coil Motion	
Appearance	1. Improper ignition coil bracket installation 2. Improper ignition coil harness installation 3. Improper ignition coil installation to bracket
Cause	Improper installation of ignition coils or missing hardware leads to premature malfunction of ignition coil extensions.
Action	Check the ignition coil and bracket for proper installation. Inspect ignition coil extensions for cracking. Reference Figure 16.

Figure 4.

Figure 4, Broken Connector Guide Tab	
Appearance	One or more connector guide tabs missing or damaged
Cause	Mishandling or improper removal
Action	Reuse ignition coil; function not affected.

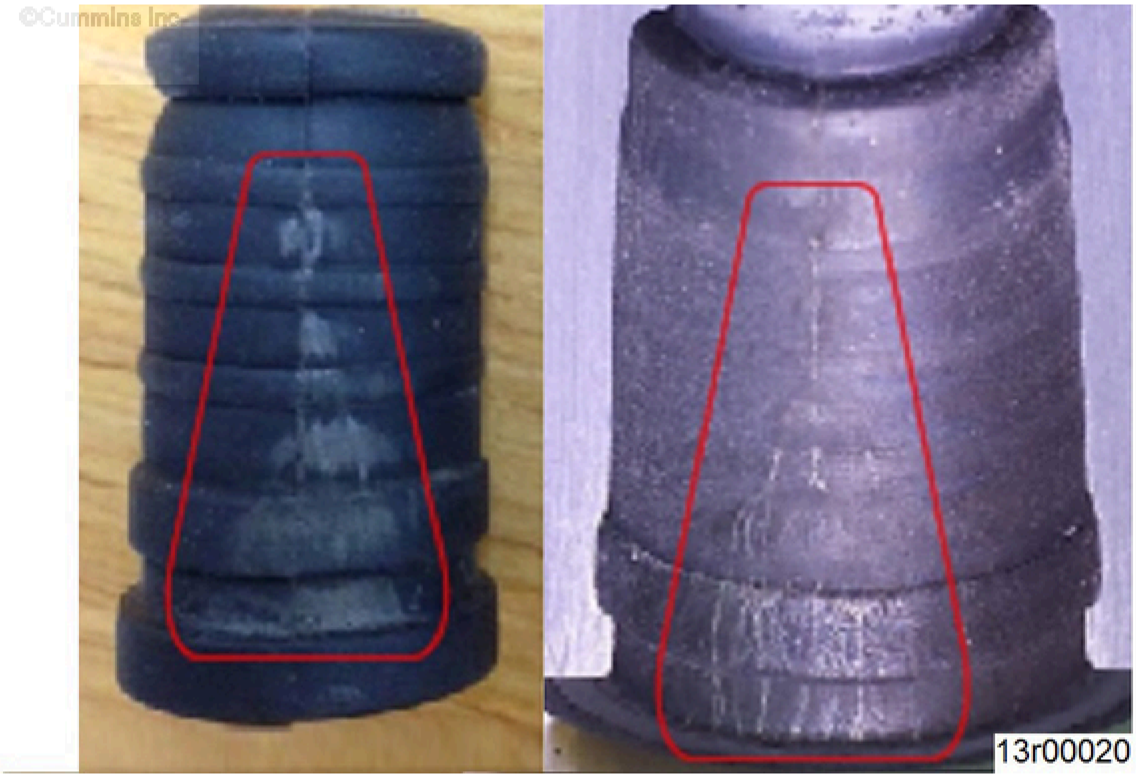

Figure 5.

Figure 5, Electrical Tracking between Boot Adapter and Spark Plug Boot	
Appearance	Tracking or “treeing” marks visible on outer diameter of spark plug boot and inner diameter of spark plug boot adapter.
Cause	Presence of an open electrical path between spark plug boot and boot adapter.
Action	Replace ignition coil extension.

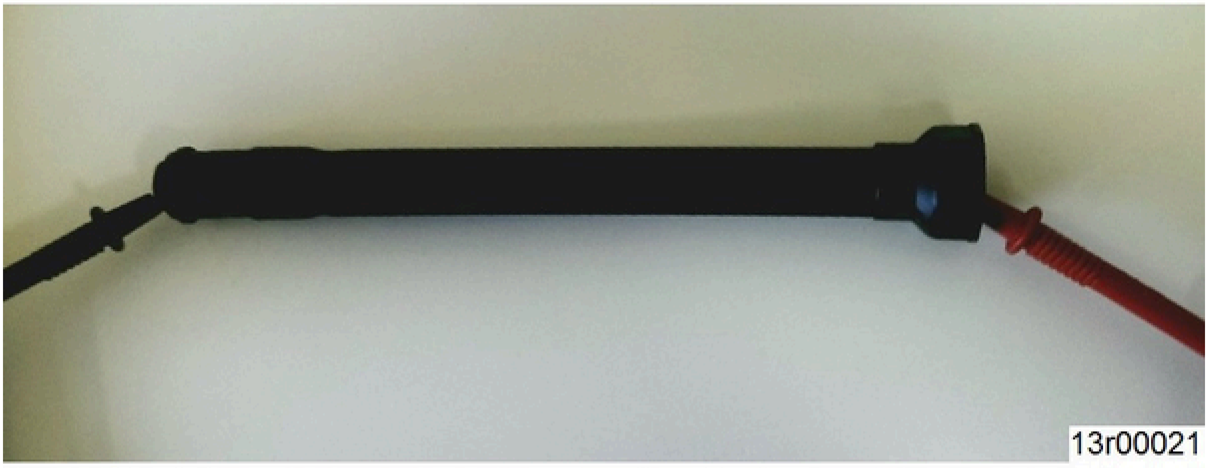

Figure 6.

Figure 6, High or Erratic Ignition Coil Extension Resistance	
Appearance	No visual indications. Measure the resistance of the ignition coil extension when detached from the ignition coil.
Cause	Loose, damaged, or corroded internal connection.
Action	If resistance is over 20k ohms, replace ignition coil extension.

Figure 7.

Figure 7, High or Erratic Spark Plug Wire Resistance	
Appearance	No visual indications. Measure the resistance of the ignition coil extension when detached from the ignition coil.
Cause	Loose, damaged, or corroded internal connection.
Action	If resistance is not within 1300 to 2200 ohms, replace spark plug wire.

Figure 8.

Figure 9.

Figure 8 and 9, Arcing Along Inner Circumference of Ignition Coil Extension	
Appearance	Etched through the spark plug boot adapter
Cause	Arcing through or inside the boot adapter
Action	Replace ignition coil extension.

Figure 10.

Figure 10, Arcing Between Ignition Coil Extension and Coil Adapter	
Appearance	Etched through coil nut
Cause	Ignition coil extension undertightened, low dielectric seal
Action	Replace ignition coil. Properly tighten ignition coil extension.

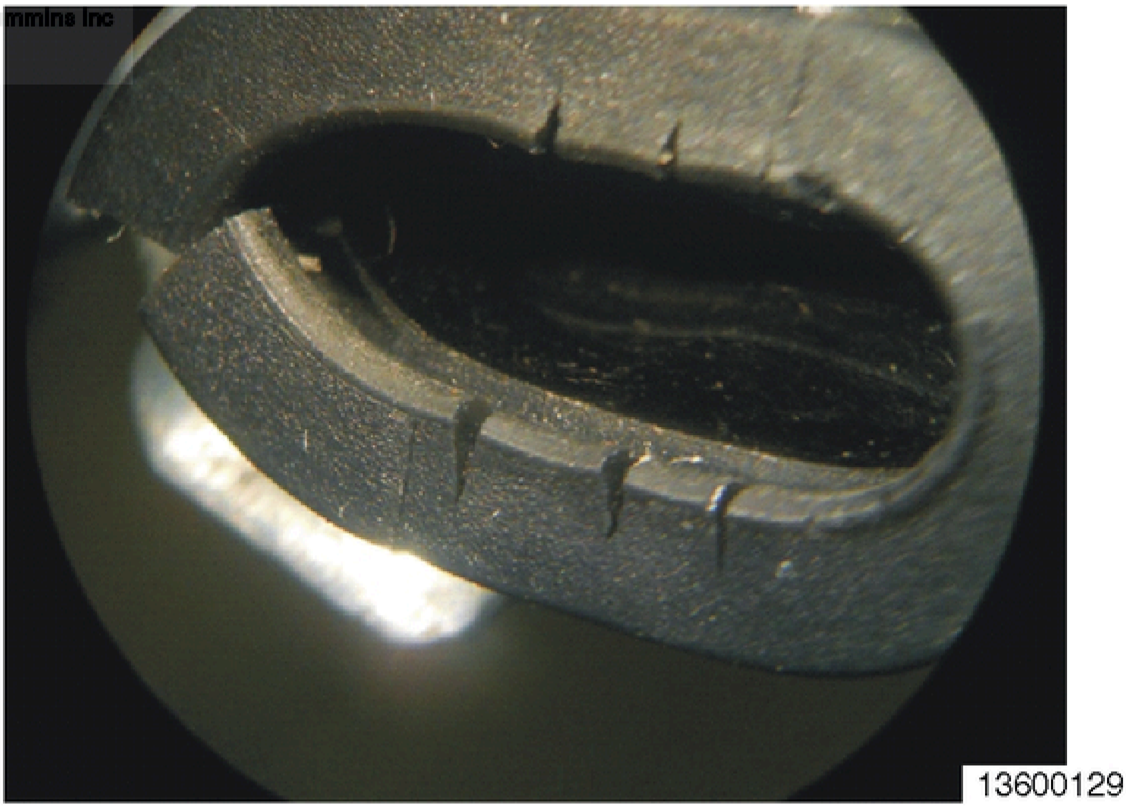

Figure 11.

Figure 11, Cracked Spark Plug Boot	
Appearance	Cracks at the end of the spark plug boot
Cause	Heat cycling, typical of a spark plug boot at the end of its useful life.
Action	Replace spark plug kit.

Figure 12.

Figure 12, Torn Spark Plug Boot	
Appearance	Tear in the spark plug boot
Cause	Mishandling when attempting to remove thespark plug boot from the ignition coil extension.
Action	Replace spark plug kit. Twist, bend, and pull when removing spark plug boot from spark plug.

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Figure 13

Figure 13, Ignition Coil Extension Damaged and Pulled Out of Ignition Coil Adapter	
Appearance	Separated coil nut and ignition coil extension tube
Cause	Spark plug boot stuck on spark plug, requiring excessive force to remove coil
Action	Replace ignition coil extension.

Left: Figure 14. Right: Figure 15.

Figure 14 and 15, Missing Spring	
Appearance	Spring is **not** present; Figure 14 - Bottom view, Figure 15 - Top view
Cause	Mishandling, spring lost during service
Action	Replace ignition coil extension. Do **not** attempt to reinstall a spring into the coil ignition coil extension.

Figure 16.

Figure 16, Broken Ignition Coil Extension	
Appearance	Ignition coil extension cracked or broken
Cause	Ignition coils removed while installed on coil bracket
Action	Replace ignition coil extension. Remove ignition coils individually. Position to the top of the bracket slots during assembly.

Figure 17.

Figure 17, Cracked Ignition Coil Extension	
Appearance	Ignition coil extension cracked or broken
Cause	Quality defect accelerated by mishandling, improper ignition coil installation, improper ignition coil bracket installation. Crack along casting weld line(s) on ignition coil extension, Figure 17 shows two cracks 180 degrees across from each other.
Action	Replace ignition coil extension. Remove ignition coils individually. Position coil to the top of the bracket slots during reassembly.

Figure 18.

Figure 18, Broken Connector or Connector Pins	
Appearance	Damaged connector pins or connector housing
Cause	Mishandling or improper removal
Action	Replace ignition coil.

Figure 19.

Figure 19, Excessive Dielectric Grease	
Appearance	Excessive amount of dielectric grease around boot and spring
Cause	Incorrect application of dielectric grease, grease buildup in spark plug boot
Action	Remove grease deposits and reuse. New boots do **not** require any grease to be added.

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Figure 20.

Figure 20, Spring Pulled Out	
Appearance	Spring protruding from spark plug boot
Cause	Spring stuck on spark plug during removal
Action	Replace ignition coil extension.

Left: Figure 21. Right: Figure 22.

Figure 21 and 22, Wrong Spark Plug Boot - ISL G	
Appearance	Larger, orange spark plug boot; Figure 21 - Wrong spark plug boot, Figure 22 - Correct boot for ISL G
Cause	Spark plug kit ordered for C Gas engine
Action	Replace with spark plug kit for ISL G.

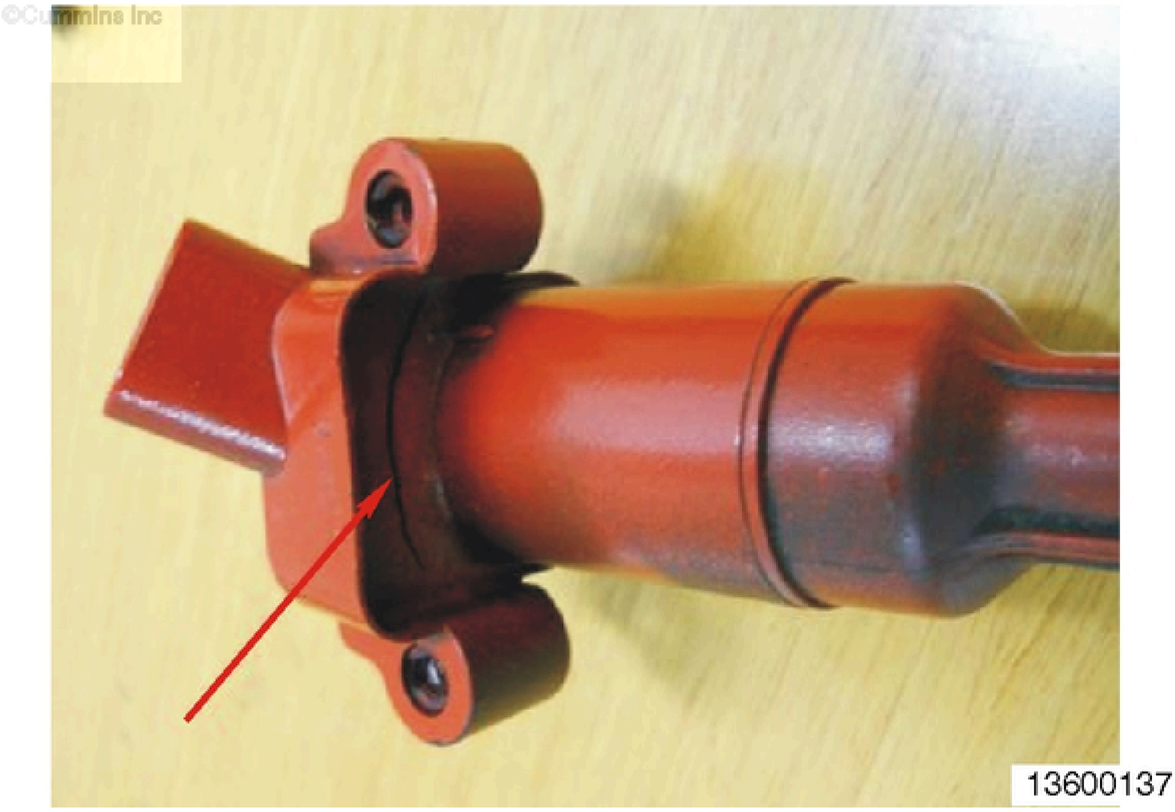

Figure 23.

Figure 23, Cracked Potting	
Appearance	Fractured underside at top of ignition coil
Cause	Ignition coil bracket out of specifications, ignition coils removed while on bracket
Action	Replace ignition coil. Remove ignition coils individually. Position to the top of the bracket slots.

Figure 24.

Figure 24, Damaged Housing	
Appearance	Chipped, cracked, or dented housing
Cause	Mishandling
Action	Replace ignition coil **only** if ignition coil fails any ignition coil troubleshooting steps.

Figure 25.

Figure 25, Ignition Coil Extension Cracked	
Appearance	Crack in ignition coil extension
Cause	Overtightened ignition coil extension
Action	Replace ignition coil extension. Properly tighten ignition coil extension.

Figure 26.

Figure 26, Ignition Coil Extension Cross Threaded	
Appearance	Gap between ignition coil extension and coil
Cause	Improper assembly
Action	**Only** replace ignition coil extension if there is no damage to the ignition coil. Replace ignition coil if threads are damaged.

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Figure 27.

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Figure 28.

Figure 27 and 28, Punctured Ignition Coil Extension	
Appearance	Small, discolored spot on the ignition coil extension
Cause	Excessively large spark plug gap

Figure 27 and 28, Punctured Ignition Coil Extension	
Action	Replace spark plug kit and ignition coil extension.

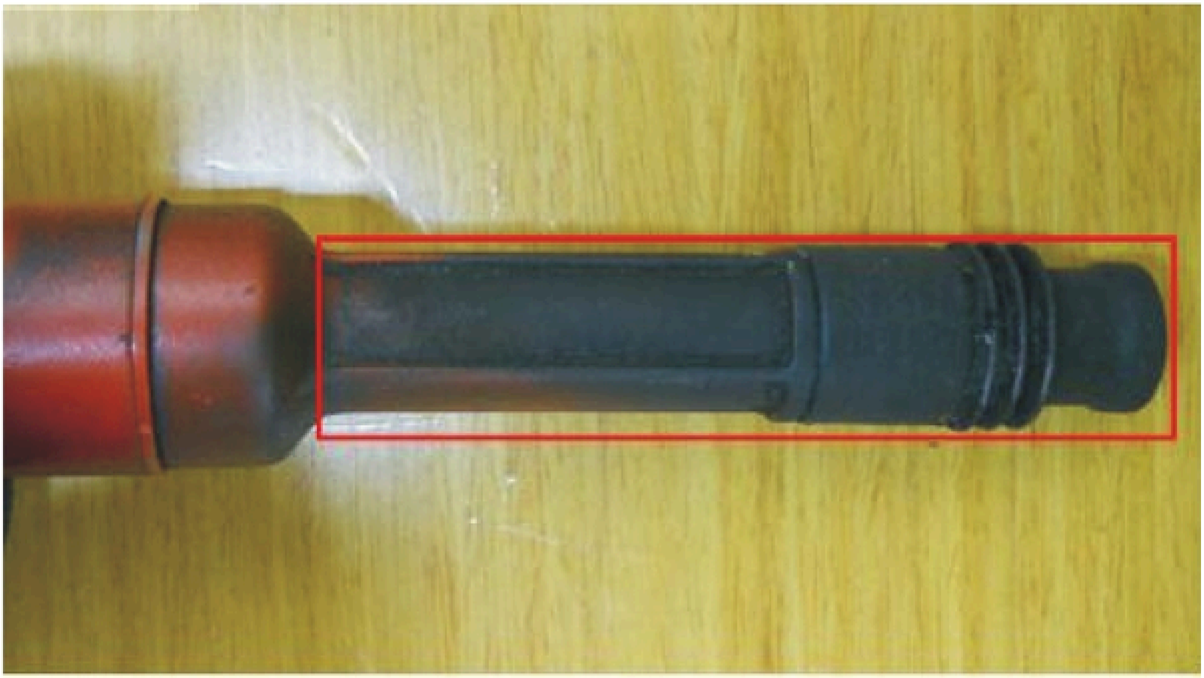

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Figure 29.

Figure 29, Bent Ignition Coil Extension	
Appearance	The ignition coil extension is slightly bent
Cause	Bent coil mounting bracket
Action	Replace ignition coil extension. Replace coil mounting bracket.

Figure 30.

Figure 30, Corroded	
Appearance	Ignition coil or ignition coil extension has yellow/orange deposit.
Cause	Water intrusion
Action	Clean deposit using an approved cleaner. Inspect grommet for cracks, tears, or cuts. Inspect valve cover sealing surface. Reference Service Manual guidelines to check ignition coil and ignition coil resistance. Reuse ignition coil and ignition coil extension if within specification. Inspect ignition coil top epoxy.

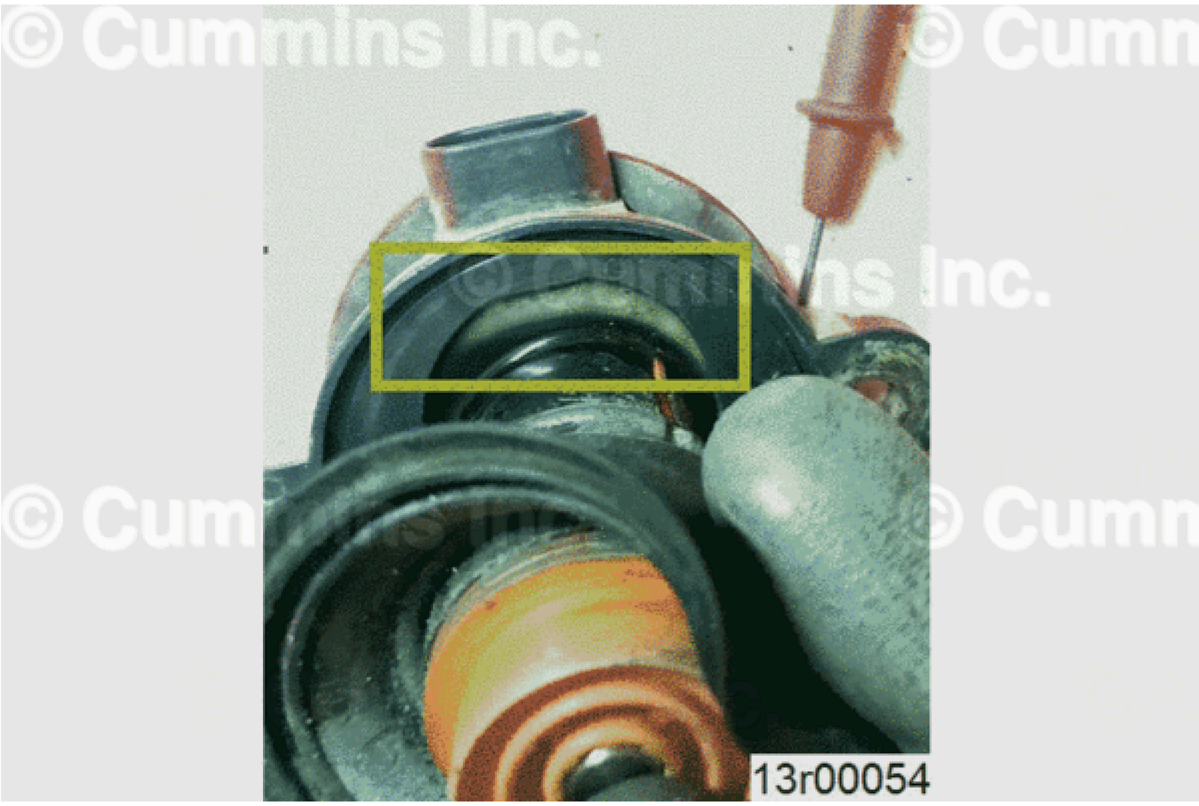

Figure 31.

Figure 31, Ignition Coil Epoxy Color Change (ISX12 G, ISX12N, ISB6.7 G, B6.7N)	
Appearance	Ignition coil epoxy potting color has changed from black to white.
Cause	Moisture/water intrusion
Action	Replace ignition coil. Reuse ignition coil extension if reuse guidelines have been met.

Figure 32.

Figure 32, Split Heat Shrink (ISX12 G)	
Appearance	Cracking on heat shrink at tip of ignition coil extension.
Cause	Fatigue due to thermal cycling can result in arcing through ignition coil extension.
Action	Replace ignition coil extension.

Figure 33.

Figure 33, Reuse Guideline for Boot Adapter (B6.7N)	
Appearance	White spark plug boot adapter when new. May appear stained and/or discolored with use.
Cause	Multiple service intervals can cause staining of the white material.
Action	Unconditional reuse, do **not** replace for discoloration or stains.

Figure 34.

Figure 34, Non-Cummins® Parts	
Appearance	Different branding; **not** a Cummins® part.
Cause	Lower voltage output causing short spark plug life.
Action	Replace ignition coil with Cummins® part.

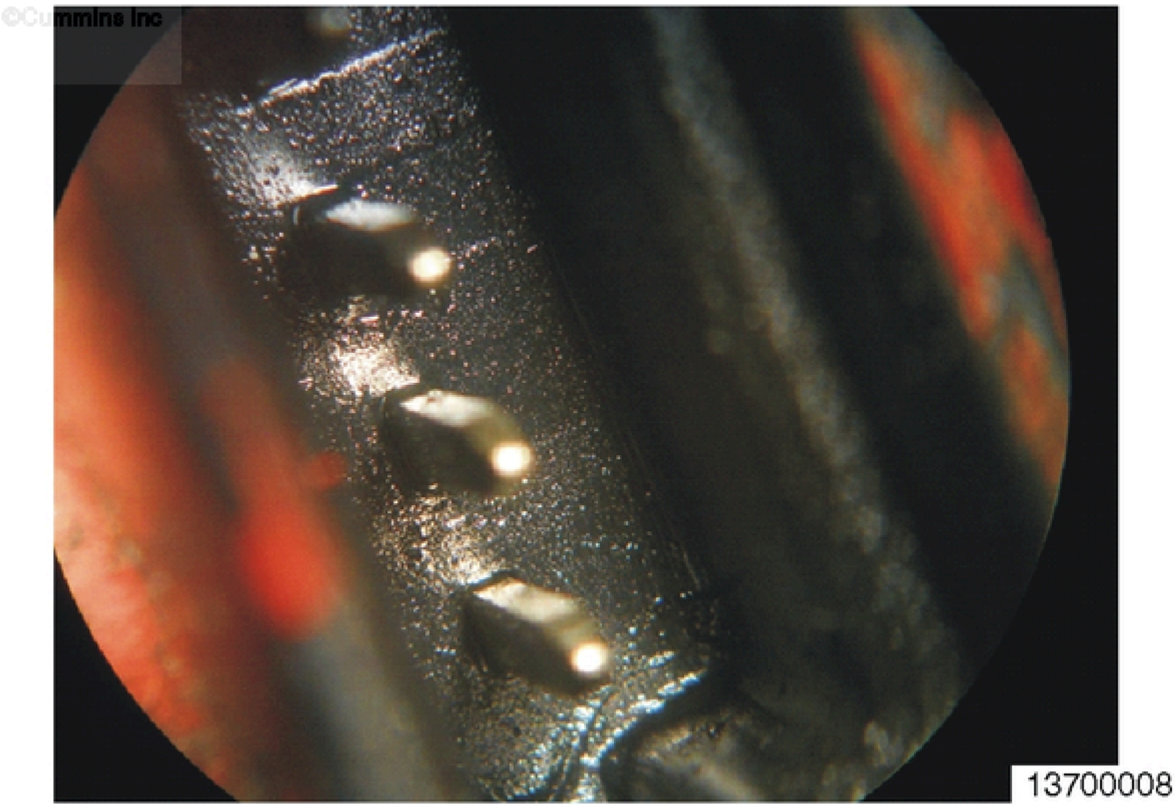

Figure 35.

Figure 35, Fluid Leak at Pin Connector	
Appearance	Wet, oily substance in the connector
Cause	Cracked coil allowing insulation fluid to leak
Action	Replace ignition coil.

Figure 36.

Figure 36, Seal Material Transferred into the Connector	
Appearance	Sticky residue from degraded connector seal or sealant
Cause	Application of incompatible chemical cleaner, adhesive, or sealant on electrical connector seal
Action	Clean residue using a compatible cleaner. Inspect electrical connector seal and locking tabs on ignition harness.

Document History

Date	Details
2014-9-3	Module Created
2014-12-18	none
2015-2-20	Added Broken Bumper image and table.
2017-3-30	Added Corroded and Split Heat Shrink (ISX12 G) and Non-Cummins® Parts tables.

Date	Details
2017-12-14	Added Top Epoxy Color Change - (ISX12 G, ISX12N , ISB6.7 G, B6.7N) table.
2020-11-16	Changes to content due to part changes. (non-greased plug boot)
2022-9-19	Added Figure 28 and added Figure numbers for all Figures.
2024-6-18	Doing check of service bulletin, none of the content was changed.
2024-9-24	Added Figure 1 and Table 1.
2025-3-27	Added new Figure 1 and Figure 2 and two types of ignition coils.

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